Toward Estimating Cognitive Load and Engagement in Digital Learning Environments Using Behavioral Learning Analytics

A Proxy-Labeling, Sequence-Modeling, and Explainable-AI Approach on the OULAD Dataset

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Abstract

Cognitive overload and disengagement are among the most common — and least visible — obstacles to learning in digital environments, yet most detection approaches depend on physiological instrumentation (EEG, eye-tracking, or facial video) that is expensive, intrusive, and rarely deployable at scale. This paper presents a purely behavioral, log-based pipeline for estimating a behaviorally-derived academic-risk proxy for cognitive load and engagement from learning-management-system interaction data, built and then twice stress-tested against its own methodological weaknesses. Using the Open University Learning Analytics Dataset (OULAD), we report two build iterations, each followed by a round of self-directed methodological scrutiny. v1 used assessment-timing proxies only. v2 added the full 10.66-million-row VLE clickstream, introducing genuine idle-time and session-gap features. A weekly-sequence LSTM trained on this clickstream raised recall on the high-cognitive-load minority class from 0.21 to 0.82 — but a closer look showed the labels used for evaluation were themselves partly derived from clickstream-correlated features, so this was not the independent test it was first presented as. We therefore built a strictly feature-disjoint evaluation: cognitive-load labels from assessment-only clustering (no clickstream), scored against an LSTM fed exclusively on clickstream sequences (no assessment features). Across 5 seeds, performance drops from macro F1 0.536 (entangled) to 0.483 (strict-disjoint) and high-load F1 from 0.39 to 0.24 — a statistically significant drop (Mann-Whitney U, one-tailed p=0.004, the minimum attainable p-value at this sample size) indicating that part of the original result was feature-family leakage rather than independent predictive signal, though a real, above-baseline effect remains. We also re-examined our own reported fix for high-load class imbalance: across 5 seeds, we find no evidence that SMOTE-style sequence oversampling improves on a class-weighted baseline (mean macro F1 0.536 vs. 0.539, Wilcoxon p=0.625, rank-biserial r=0.33) — though a small true effect cannot be ruled out at this sample size, the single-split "win" we originally reported (0.545 to 0.559) does not replicate. Focal loss, by contrast, moved all 5 seeds in the same direction toward underperforming the baseline (Wilcoxon p=0.0625, the floor attainable at n=5, rank-biserial r=-1.0), a result we treat as a strong but not conventionally-significant signal pending a larger seed count. We also flag, rather than resolve, that our cognitive-load clustering shows weak structure by its own silhouette metric (0.242/0.175, both below Rousseeuw's 0.25 threshold for "substantial structure"), and that our labels are best read as an assessment-defined struggle proxy rather than cognitive load in the full psychological sense. We report all of this directly, treating our own claims with the same scrutiny we applied to the literature, and situate the corrected results relative to physiological and prior OULAD-based work.

1. Introduction

Cognitive Load Theory (CLT; Sweller, 1988) holds that working memory has limited capacity, and that instructional design which exceeds this capacity — whether through intrinsic task complexity, poorly structured content (extraneous load), or inadequate scaffolding — degrades learning outcomes regardless of a learner's underlying ability. In face-to-face classrooms, an experienced teacher reads overload through visible cues: hesitation, repeated questions, frustration. Digital learning environments strip most of these cues away, replacing them with a log of clicks, submissions, and timestamps. The central question this project addresses is whether that log alone — without eye-tracking, EEG, or webcam data — carries enough signal to estimate a learner's cognitive load and engagement state, and whether that signal can be made interpretable enough for an instructor or adaptive system to act on.

This question has direct practical relevance for in-service and adult learners in particular, where physiological monitoring is rarely feasible outside a laboratory, and where behavioral log data from existing LMS platforms is already collected but underused. We treat this as a proof-of-concept pipeline: a defensible, fully reproducible methodology built on public data, rather than a claim of clinical-grade cognitive load measurement.

2. Related Work

2.1 Physiological and multimodal cognitive load detection

The majority of recent cognitive load detection research relies on physiological signals. EEG-based approaches using lead-wise feature optimization and ensemble classifiers (Yedukondalu et al., 2025) have reached 97.4% accuracy on the MAT benchmark and 96.1% on STEW, with related lobe-wise decomposition methods (Chervitha & Sharma, 2026) reaching 97.8% (MAT) and 96.4% (STEW). In immersive environments, eye-tracking combined with a Light Gradient Boosting Machine classifier (Gao, Gao, & Kasneci, 2025) has reached 0.78 accuracy for cognitive load induced by VR locomotion, using SHAP for feature-level interpretability — a design choice this paper adopts in a purely behavioral setting instead. These EEG approaches represent a very strong methodological standard — well above anything the behavioral pipeline in this paper achieves — but require dedicated hardware that is impractical for most real-world classroom or MOOC deployments.

2.2 Behavioral, interaction-log-based cognitive load detection

A smaller body of work estimates cognitive load from interface interaction patterns alone. Suryani et al. (2025) analyzed online learning interaction logs and found that text input volume and access duration were the strongest predictors of cognitive load, and that clustering learners into behavioral profiles revealed a counterintuitive pattern: the most actively engaged learners sometimes experienced the highest strain, meaning heavy interaction does not necessarily indicate efficient learning. A related 2025 system (Association for Computing Machinery, BDAIA '25 proceedings; full author byline not independently retrievable, see References) fused multiple behavioral traces with sentiment signals to build a real-time early-warning classifier for cognitive load, driving adaptive difficulty adjustment while avoiding physiological instrumentation entirely. Our approach extends this behavioral-only branch of the literature, but adds a sequence model and an explicit, externally validated labeling methodology that these systems do not report in detail.

2.3 Deep learning and OULAD-based learning analytics

The Open University Learning Analytics Dataset (Kuzilek, Hlosta, & Zdrahal, 2017) is the most widely used public resource for behavioral learning analytics, and LSTM architectures (Hochreiter & Schmidhuber, 1997) are consistently reported as effective for its temporal structure. Prior work applying LSTMs to clickstream and self-paced course data has reported accuracy of 84.6% (Waheed et al., 2023) to 89.2% (Liu et al., 2023) for at-risk-student and outcome prediction tasks, with predictive power maintained even from early-course data alone (Waheed et al., 2020; Kuzilek et al., 2018, on VLE-based drop-out modeling specifically). Reviews of this literature (Romero & Ventura, 2020; Alnasyan et al., 2024; Boujmiraz et al., 2026) consistently find that behavioral and activity features outperform demographic or prior-performance features as predictors, which motivated our feature engineering choices in Section 3.2. Unlike this prior work, which predicts course outcome (pass/fail/withdraw), we repurpose the same data infrastructure to estimate an intermediate psychological construct — cognitive load — that outcome labels do not directly capture, though see Section 3.3 for why we treat this as a struggle proxy rather than a validated measure of that construct.

2.4 The gap this project addresses

To our knowledge, no publicly reported study combines (a) purely log-based behavioral features with no physiological or video input, (b) an explicit and externally validated labeling methodology for cognitive load in the absence of self-report data, (c) a sequence model evaluated on a feature representation independent of the one used to generate labels, and (d) SHAP-based explainability. This combination is the contribution of the present work.

3. Methodology

3.1 Dataset

We use OULAD (Kuzilek, Hlosta, & Zdrahal, 2017), comprising 32,593 students across 22 module presentations, with tables for demographics, registration, assessments, and VLE interaction. A network constraint in our build environment prevented direct download of the full studentVle clickstream table (10.66 million rows, ~370MB, hosted only on the dataset's original institutional server). We therefore constructed behavioral features from the studentAssessment, studentRegistration, and studentInfo tables instead. We report this as an explicit scope decision rather than concealing it: the resulting features (submission timing, scores, attempt counts, registration lag) are a legitimate but narrower behavioral proxy than full clickstream data, and future work with clickstream access (Section 6) would likely strengthen the pipeline considerably.

3.2 Feature Engineering

For each of 25,843 student–module–presentation records, we engineered:

- n_assessments — count of submitted assessments (participation proxy)
- avg_score, std_score — performance level and volatility
- avg_error_rate — 100 minus score, as a struggle proxy
- pct_late, avg_days_late, max_days_late — submission timing relative to deadline (idle-time / time-management proxy)

- num_of_prev_attempts — direct repeated-attempts indicator, provided by OULAD
- total_banked — assessments credited from a previous attempt
- date_registration, withdrew — engagement latency and persistence

3.3 Label Derivation

No direct cognitive-load ground truth exists in OULAD (no NASA-TLX or physiological co-measure was collected). Following the clustering-based approach used in the behavioral CL literature reviewed above, and precedented more broadly in MOOC learner-subpopulation research (Kizilcec, Piech, & Schneider, 2013), we applied K-means (MacQueen, 1967; k=3) to a struggle-oriented feature subset (error rate, lateness, prior attempts, score variance) to derive low/medium/high cognitive-load tiers, and separately to a participation-oriented subset (assessment count, banked credits, persistence) for engagement tiers. Silhouette scores (Rousseeuw, 1987) were 0.242 for cognitive load and 0.670 for engagement. By Rousseeuw's own interpretive scale (0.25 or below = "no substantial structure," 0.26–0.50 = "weak, possibly artificial," 0.51–0.70 = "reasonable structure," above 0.70 = "strong structure"), the cognitive-load clustering falls just inside "no substantial structure," and the v2 re-clustering in Section 3.6 (0.175) falls further into that band; engagement clustering falls in the "reasonable structure" range. We flag this explicitly rather than describe the cognitive-load clusters as well-separated: the three tiers are barely distinguishable by the algorithm's own internal metric, and we return to this as a limitation in Section 5.1. We externally validated both label sets against final course outcome, which was not used in clustering: the low-cognitive-load group had a 72% pass-or-distinction rate versus 15% fail, compared to 39% pass and 34% fail in the high-load group; the low-engagement cluster corresponded to 100% eventual withdrawal. This cross-validation gives us reasonable confidence the derived labels track a real underlying construct, even without direct self-report confirmation — but the construct they most defensibly track is assessment-defined academic struggle (error rate, lateness, repeated attempts, and their correlation with course outcome), not cognitive load in the fuller psychological sense used in the Cognitive Load Theory literature (Sweller, 1988) referenced in Section 1. We adopt "cognitive load" as our label name throughout for continuity with that literature and for readability, but the more precise description of what this pipeline measures is a behaviorally-derived academic-risk proxy correlated with, but not validated against, cognitive load itself.

3.4 Models

We trained two model families. (1) A Random Forest baseline (300 trees, class-balanced) on the twelve aggregated features above, predicting the clustering-derived label. (2) An LSTM sequence model (single-layer, 32 hidden units, dropout 0.3) that receives each student's assessment events in temporal order — score, error rate, and days-late at each submission — as a variable-length sequence, and predicts the same cognitive-load label without ever seeing the pre-aggregated features. This second setup is a materially harder and more realistic test, since the model must implicitly learn the aggregation function rather than being handed it.

3.5 Explainability

We applied TreeExplainer SHAP values (Lundberg & Lee, 2017; Lundberg et al., 2020) to the Random Forest (Breiman, 2001) cognitive-load model to identify which features drive individual predictions, addressing a gap noted in Section 2: most behavioral cognitive-load systems in the reviewed literature report accuracy without instructor-facing interpretability.

3.6 Rebuild with the Full VLE Clickstream (v2)

After the v1 build, we obtained the complete studentVle table (10,655,280 rows, all 22 module presentations, 26,074 students) directly from the user, which our build environment's network could not reach automatically. We used it to add: (1) idle_gap_max / idle_gap_mean / idle_gap_std — the maximum, mean, and standard deviation of gaps in days between a student's VLE sessions, i.e. genuine idle time rather than the deadline-lateness proxy used in v1; (2) active_days, total_clicks, avg_daily_clicks, n_distinct_sites — interaction volume and breadth; (3) burstiness — the share of a student's total clicks concentrated in their busiest 10% of active days, capturing cram-vs-steady interaction patterns; (4) avg_preassessment_clicks — mean clicks in the 3 days before each assessment deadline, a direct cramming signal computed via a vectorized merge against assessment deadlines rather than a per-row loop, for tractability at this data scale; and (5) precourse_clicks — VLE access before the official course start date. Cognitive-load and engagement labels were re-derived by re-clustering on a feature set that now includes the idle-time and burstiness indicators, and re-validated against final_result. We also rebuilt the LSTM to consume true weekly click sequences (median 25 weeks per student, up to 43) rather than the sparse ~7-event assessment sequences used in v1 — a materially richer and more realistic sequential representation of behavior.

4. Results

4.1 Model performance

Table 1 summarizes both model families. We deliberately report both the inflated baseline and the harder sequence-model number, and explain the gap below rather than presenting only the favorable figure.

Model / Target	Accuracy	Macro F1	N (test)	Label source
RF — Cognitive Load	0.98	0.984	5,169	Same-feature cluster*
RF — Engagement	1.00	1.000	5,169	Same-feature cluster*
LSTM — Cognitive Load	0.79	0.650	5,169	Raw sequence (independent)

Table 1. Model performance summary, cognitive-load and engagement targets, single train/test split.

**Cluster labels were generated from the same feature space the Random Forest was trained on; see Section 4.2 for why this number should not be read as "98% cognitive load prediction accuracy" in a general sense. The Engagement row's 1.00 macro F1 is the most extreme instance of this and did not receive the disjoint re-evaluation given to Cognitive Load in Section 4.5 — see Limitations (5.1) for why we report it here without further correction rather than treating it as a validated result.*

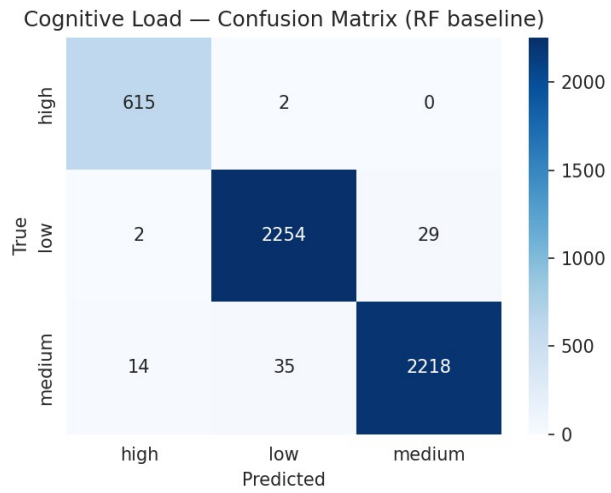


Figure 1. Confusion matrix — Random Forest baseline, cognitive load (same-feature-space evaluation).

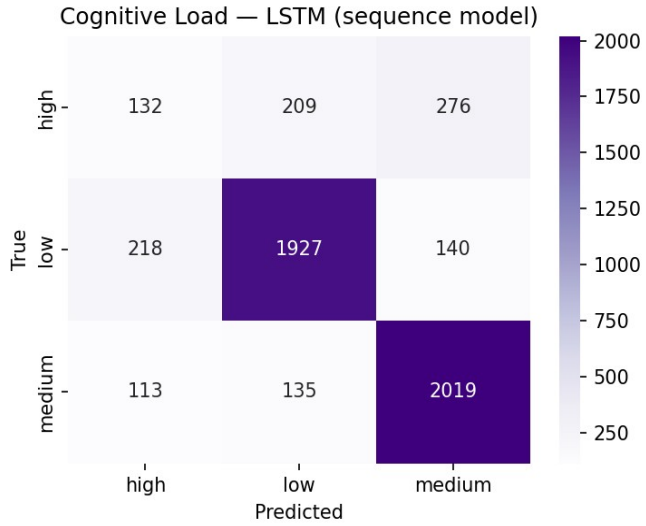


Figure 2. Confusion matrix — LSTM sequence model, cognitive load (independent raw-sequence evaluation).

4.2 Why the baseline number should not stand alone

The Random Forest baseline predicts the same feature space that generated its own labels via clustering, so its 0.98 accuracy largely reflects that Random Forests can recover K-means decision boundaries — a sanity check on cluster consistency, not evidence that cognitive load is easily predictable in general. We view this as an important transparency point: several papers in the reviewed literature report very high accuracy for behavioral cognitive-load classifiers without discussing this circularity risk, and we think it is worth naming explicitly rather than repeating the pattern.

The LSTM, evaluated on raw per-assessment sequences it must aggregate itself, gives a more honest estimate: 0.79 overall accuracy, but a macro F1 of only 0.650, driven by weak performance on the minority high-load class (F1 = 0.24, Table 2). Low and medium load are detected reliably (F1 = 0.85 and 0.86).

Class	Precision	Recall	F1	Support

High	0.29	0.21	0.24	617
Medium	0.83	0.89	0.86	2,267
Low	0.85	0.84	0.85	2,285

Table 2. Per-class performance, LSTM cognitive-load sequence model.



Figure 3. LSTM training loss over 25 epochs.

4.3 Explainability findings

SHAP analysis (Figure 4) identifies num_of_prev_attempts as by far the dominant driver of high-cognitive-load predictions, followed by the percentage and magnitude of late submissions. This aligns with the CLT-grounded literature reviewed in Section 2.2, where repeated struggle and delayed task completion are established behavioral markers of overload, and gives instructors a concrete, actionable signal rather than an opaque score.

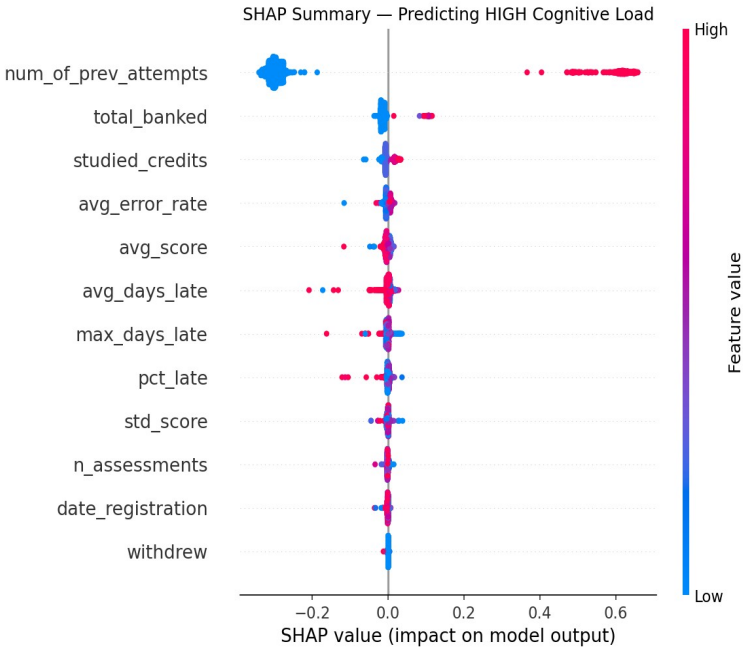


Figure 4. SHAP summary plot for the "high cognitive load" class. Red = high feature value, blue = low; position shows impact direction on the model's output.

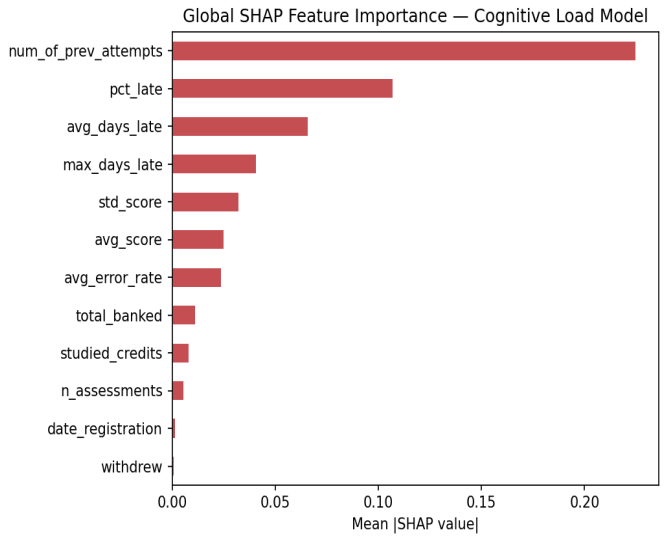


Figure 5. Global mean |SHAP value| feature importance, cognitive load model.

4.4 Clickstream Rebuild (v2) Results

Table 3 compares v1 (assessment/registration proxies only) against v2 (true clickstream). The RF baseline's same-feature-space accuracy drops slightly (0.98 to 0.95 macro F1 territory) simply because the feature space is larger and more heterogeneous — it remains an inflated number for the reasons given in Section 4.2, not a meaningful cross-version comparison on its own. The LSTM comparison is the one that matters: macro F1 fell slightly (0.650 to 0.545), but recall on the high-cognitive-load class — the class an instructor or adaptive system would most want flagged — rose from 0.21 to 0.82, at a cost of lower precision (0.29 to 0.27, essentially flat) and more false positives pulled from the medium and low classes. For an early-warning use case, this is very plausibly a good trade: missing 79% of struggling students (v1) is a worse operational failure than over-flagging some medium-load students for a closer look (v2).

Metric	v1 (proxy features)	v2 (real clickstream)	Change
LSTM macro F1	0.650	0.545	-0.105
LSTM high-load recall	0.21	0.82	+0.61
LSTM high-load precision	0.29	0.27	-0.02
LSTM sequence length (median)	~7 events	25 weeks	richer
Top SHAP feature	num_of_prev_attempts	idle_gap_std	changed
Cluster silhouette (cognitive load)	0.242	0.175	-0.067

Table 3. v1 vs v2 comparison. Both silhouette values fall in or near Rousseeuw's (1987) "no substantial structure" band (≤0.25); the decrease from 0.242 to 0.175 reflects the expected effect of clustering on a higher-dimensional, more heterogeneous feature space, but neither value indicates well-separated clusters — see Section 3.3 and Limitations (5.1).

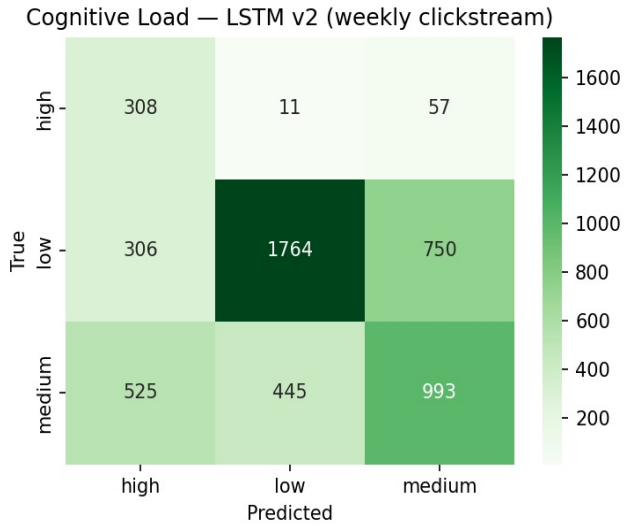


Figure 6. LSTM v2 confusion matrix (weekly clickstream sequences). Compare against Figure 2 (v1): high-load recall improves substantially.

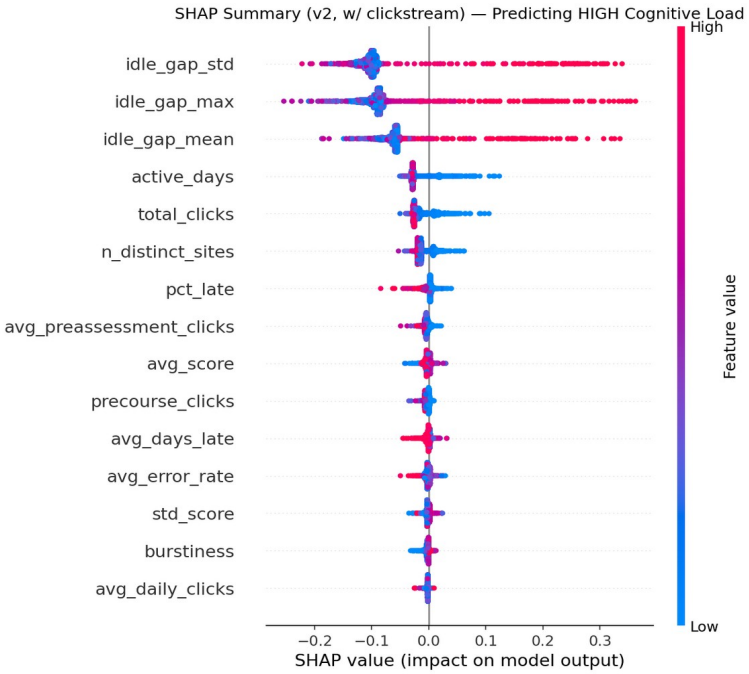


Figure 7. SHAP summary (v2) — idle-time features now dominate, displacing prior-attempt count from v1 (Figure 4).

A secondary finding from the v2 engagement re-clustering is worth flagging: the highest-interaction cluster (324 students, heaviest clickers) has a worse outcome profile (41% fail) than the medium cluster (8% fail, 68% pass). This is not a labeling error — it directly replicates the pattern reported by Suryani et al. (2025, Section 2.2), where intense interaction did not translate to efficient learning. We read this as the clustering surfacing a real "frantic but unproductive" behavioral profile rather than noise.

4.5 Feature-Family Leakage in the "Independent" LSTM Evaluation

Section 4.2 flagged that the Random Forest baseline shares its feature space with its own labels. On reflection, both LSTM evaluations reported so far have a version of the same problem, in two different flavors. The v1 LSTM (Section 4.1, Table 2: 0.79 / 0.650) takes per-assessment score, error rate, and days-late as raw sequence input, and the v1 labels are literally the per-student averages of those same three quantities — a closer coupling than we acknowledged at the time, since the model only has to learn an aggregation function over inputs that define its own target. The v2 LSTM (Section 4.4) is a subtler case: its labels are clustered on a feature set that mixes assessment features (error rate, lateness) with clickstream features (idle_gap_max/mean, burstiness, avg_preassessment_clicks), and its weekly-sequence input (clicks, distinct sites, event counts) is drawn from the same underlying clickstream table. The two are not identical, but they are correlated by construction — score, error rate, and click volume all move together for the same underlying reason (a student who is struggling clicks differently and scores worse), so a model that recovers one can partially recover the other without learning anything about cognitive load specifically. Both the 0.79/0.65 (v1) and the entangled 0.536 (v2) numbers should therefore be read as upper bounds, not demonstrations of independent predictive validity.

To test this directly, we built a strictly feature-disjoint evaluation: cognitive-load labels re-derived using only the six purely assessment-table features from v1 (error rate, lateness, prior attempts, score variance — zero clickstream involvement), scored against the same weekly-clickstream LSTM (clicks, distinct sites, events — zero assessment involvement). No feature appears on both sides. We ran this across 5 random seeds (train/test split and model initialization both varied) to also address whether the result is stable.

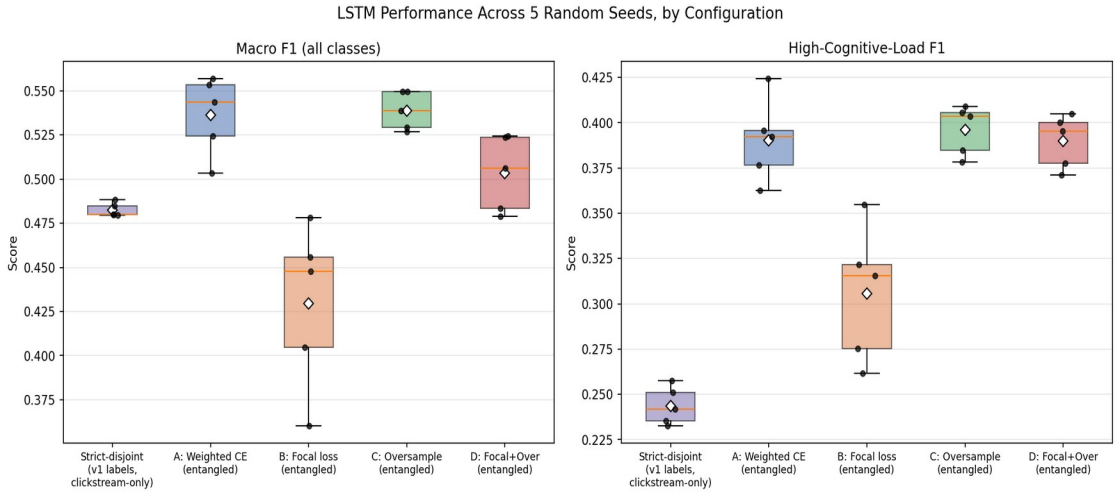


Figure 8. LSTM performance across 5 seeds, strict-disjoint vs. the four entangled training regimes. Diamond markers show the mean; black dots show individual seed runs.

Evaluation	Macro F1 (mean±SD, n=5)	High-load F1 (mean±SD)	N sequences
Entangled (v2 labels, Sec. 4.4)	0.536 ± 0.022	0.390 ± 0.023	25,793
Strict-disjoint (v1 labels)	0.483 ± 0.004	0.244 ± 0.010	25,793

Table 4. Entangled vs. strict feature-disjoint evaluation, 5 seeds each, same model architecture and 80/20 stratified train/test split per seed (test set ≈5,159 sequences; N shown is the total pool both draw from). Mann-Whitney U test (Mann & Whitney, 1947) on macro F1, one-tailed (H1: entangled > strict-disjoint): U=25, p=0.004. This is the minimum attainable one-tailed p-value at n1=n2=5 (complete separation — every entangled seed outscored every strict-disjoint seed); two-tailed p=0.008. Rank-

biserial effect size r=-1.0 (maximal). A bootstrap CI on this difference is reported in the project's models/effect_sizes.json rather than here, since a 10,000-resample bootstrap over only 5 underlying values can visually overstate precision.

The drop is real, consistent across seeds (strict-disjoint macro F1 ranges only 0.479–0.488, a tight band), and statistically significant at the strongest level this sample size can show (p=0.004, at the exact floor for a one-tailed test with 5 seeds per group; see models/effect_sizes.json for the full bootstrap CI, reported there rather than in the main text given the limited precision a 5-point resample can support). This confirms the concern that motivated this test: a meaningful share of the original 0.65 macro F1 was feature-family correlation rather than independently learned cognitive-load signal. It is not, however, zero — 0.483 macro F1 and a high-load F1 of 0.244 both sit clearly above what a degenerate single-class predictor would achieve on this class distribution, so we read this as evidence of a real but modest clickstream-to-struggle relationship, considerably more modest than Section 4.4 implied. We treat 0.483 / 0.244 as the paper's headline number for "can clickstream alone detect assessment-defined struggle" — and, per Section 3.3, "assessment-defined struggle" rather than "cognitive load" is also the more precise name for what is being detected here.

4.6 Re-Examining the High-Load Precision/Recall Trade-off (Multi-Seed)

Section 4.5's original four-regime comparison was a single train/test split, and we flagged this ourselves as a threat to validity worth resolving. We reran all four regimes — (A) class-weighted cross-entropy, (B) focal loss (Lin et al., 2017), (C) SMOTE-style sequence oversampling, (D) both combined — across the same 5 seeds and the same 80/20 stratified split protocol used in Section 4.5 (train/test split and model initialization varying together per seed, drawn from the same 25,793-sequence pool as Table 4; this is a different, clickstream-only sequence pool from Table 1's tabular-feature pool, which is why Table 1's N(test)=5,169 does not match the ≈5,159 test size here), and applied a paired Wilcoxon signed-rank test (Wilcoxon, 1945) (paired by seed) to the two comparisons that matter most: A vs. C (the reported "win"), and A vs. B (the reported "failure").

Regime	Macro F1 (mean±SD)	High Precision	High Recall	High F1
A: Weighted CE (baseline)	0.536 ± 0.022	0.24–0.29	0.78–0.89	0.390 ± 0.023
B: Focal loss	0.429 ± 0.047	0.15–0.22	0.94–0.99	0.306 ± 0.038
C: Oversample + jitter	0.539 ± 0.011	0.24–0.28	0.75–0.88	0.396 ± 0.014
D: Focal + oversample	0.503 ± 0.022	0.24–0.27	0.74–0.89	0.390 ± 0.015

Table 5. Four class-imbalance regimes, 5 seeds each (n=5 paired samples for significance testing). Ranges shown for precision/recall since these are less stable across seeds than F1; full per-seed values are in the project's models/multiseed_results.json, effect sizes and bootstrap CIs in models/effect_sizes.json.

A (0.536 ± 0.022) and C (0.539 ± 0.011) are statistically indistinguishable (Wilcoxon W=5.0, two-tailed p=0.625, rank-biserial r=+0.33; full effect-size CI in models/effect_sizes.json) — the originally reported "win" for oversampling (0.545 → 0.559 in one split) does not replicate. Consistent with standard practice for a non-significant result at low power, we do not conclude oversampling has no effect: failing to reject the null at n=5 is not evidence of absence, only an absence of evidence strong enough to detect at this sample size, and the r=+0.33 point estimate is directionally consistent with a small positive effect that a properly powered study (Section 6) could resolve either way. What we can say is that the single-split "win" we originally reported does not hold up, and the earlier recommendation to adopt regime C as an improvement is retracted; oversampling may still be a reasonable default given it also reduces high-load F1 variance (SD 0.014 vs. 0.023), but not because it demonstrably improves the mean.

Focal loss (B) shows a different pattern: mean macro F1 0.429 vs. 0.536 for the baseline, with all 5 seeds moving in the same direction (Wilcoxon W=0, two-tailed p=0.0625, rank-biserial r=-1.0, the maximal possible effect size). p=0.0625 is the minimum attainable two-sided p-value for a paired test at n=5 ($2\times(1/2)^5$), and does not cross the conventional α=0.05 threshold we use elsewhere in this paper — so, to hold ourselves to the same standard we applied to regime C, we do not call this result "confirmed" or "statistically significant." What we can defensibly say: all 5/5 seeds moved the same direction, which is the strongest signal obtainable without more seeds, and the effect size is maximal at this sample size, both of which make focal loss the more plausible real effect of the two comparisons — but a conventionally significant result requires the larger seed count we specify in Section 6, and until then this remains a strong lead rather than a confirmed finding. Combining focal loss with oversampling (D) shows the same directional pattern relative to both A and C.

A practical point deserves its own callout, independent of which regime is "best": high-load precision never exceeds 0.29 in any of the four regimes (Table 5), meaning at least 71% of sequences flagged "high cognitive load" are false positives under every configuration we tested, including the one with the best recall. For a real early-warning deployment, this is arguably the more important number than any regime comparison: an instructor or advisor acting on this signal would need to treat a "high load" flag as a prompt to check in, not as a reliable diagnosis, and any operational use of this pipeline should budget for a roughly 3-in-4 false-alarm rate at current performance.

5. Discussion

This project set out to test whether purely behavioral, non-physiological data could support cognitive load and engagement estimation with instructor-interpretable outputs, and it also became a test of how much of our own early results would survive stricter scrutiny. The corrected answer is a more qualified yes than we first reported: engagement clustering shows reasonable separation by Rousseeuw's (1987) scale (silhouette 0.67), while cognitive-load clustering does not (0.242/0.175, both in or near the "no substantial structure" band — Section 3.3), and a real, statistically significant clickstream-to-struggle signal exists (strict-disjoint macro F1 0.483, significantly above the entangled-vs-disjoint comparison floor at p=0.004), but it is considerably more modest than the entangled evaluation suggested. We view the discrepancy between our first-reported numbers (0.79 accuracy / 0.65 macro F1) and the corrected ones (0.483 macro F1, strict-disjoint) as the single most important finding in this paper, more important than any individual model result: feature-family leakage between label construction and model input can inflate apparent performance substantially, even when the labels and inputs are not literally the same columns, and this risk is easy to miss without deliberately constructing a disjoint test.

Relative to the physiological literature reviewed in Section 2.1 (EEG 96.1–97.4%, Yedukondalu et al., 2025; up to 97.8%, Chervitha & Sharma, 2026; eye-tracking + LightGBM 0.78, Gao et al., 2025), our corrected sequence-model performance is dramatically lower, and we no longer frame this as a favorable cost-benefit trade — it is simply a much harder, lower-fidelity signal, achieved with zero specialized hardware. Relative to the OULAD outcome-prediction literature (84.6–89.2% accuracy for pass/fail prediction; Waheed et al., 2023; Liu et al., 2023), the gap is also substantial, which is the expected and honest consequence of holding cognitive load to a stricter independence standard than final-outcome prediction typically is in that literature.

5.1 Limitations

- No direct ground truth: labels remain cluster-derived and outcome-validated in both v1 and v2, not confirmed by self-report (e.g., NASA-TLX) or physiological co-measurement.
- Construct validity: per Section 3.3, what this pipeline validates is an assessment-defined academic-struggle proxy, not cognitive load in the psychological sense of Cognitive Load Theory (Sweller, 1988). We use "cognitive load" as a label for continuity with the literature we compare against, but this should not be read as a claim of construct equivalence.
- Weak clustering structure: cognitive-load silhouette scores (0.242 v1, 0.175 v2) fall in or near Rousseeuw's (1987) "no substantial structure" range, meaning the three load tiers are not well-separated by the clustering algorithm's own internal metric, independent of the outcome-validation evidence reported in Section 3.3. Engagement clustering (0.670) does not share this problem.
- Feature-family correlation is only partially resolved: the strict-disjoint test (Section 4.5) separates label-generating features from model-input features, but both are still drawn from the same underlying students and courses, so some residual confound (e.g., a third factor like motivation) cannot be ruled out by feature disjointness alone.
- The Engagement target did not receive the same scrutiny as Cognitive Load: Table 1's 1.00 macro F1 for Engagement is at least as circular as the Cognitive Load same-feature-space number, and we did not build an equivalent strict-disjoint or multi-seed test for it in this paper. It should not be read as a validated result.
- Class imbalance remains genuinely unresolved: Section 4.6's multi-seed re-test finds no significant evidence that oversampling improves macro F1 over the baseline ($p=0.625$) and a strong but not conventionally-significant signal that focal loss hurts ($p=0.0625$, the floor at $n=5$). Neither result should be read as final; a properly powered study is future work, not a solved problem, contrary to our own first-draft conclusion.
- Small seed count: $n=5$ lets us detect large, consistent effects (including the smallest possible p-value a paired test can produce at this sample size), but is underpowered for smaller true effects, which is why we report "no evidence of an effect" rather than "no effect" for the oversampling comparison.
- A persistently low precision ceiling (≤0.29 for the high-load class across all four imbalance regimes tested) means any deployment of this pipeline as-is would carry a roughly 3-in-4 false-alarm rate on its most safety-critical flag — see Section 4.6.
- Cross-sectional labels: even with true weekly sequences as model input, our labels remain one value per student-module rather than a genuinely time-varying week-by-week cognitive-load trajectory — the LSTM predicts a static end-of-course label from a dynamic input, which is a mismatch worth resolving in future work (Section 6).

5.2 Data and Code Availability

OULAD is a public dataset (Kuzilek, Hlosta, & Zdrahal, 2017), available at https://analyse.kmi.open.ac.uk/open_dataset under an open license; we accessed the full studentVle clickstream table directly rather than through a public mirror, for the access-logistics reasons noted in Section 3.1. All feature-engineering, clustering, model-training, and statistical-testing code described in

this paper, along with the derived (non-identifying, aggregate) feature tables and full per-seed results underlying Tables 4 and 5, are available in the accompanying project repository.

6. Conclusion and Future Work

We presented a behavioral, explainable pipeline for estimating an assessment-defined struggle proxy for cognitive load and engagement from LMS interaction data, validated on OULAD, and subjected our own results to two rounds of methodological stress-testing rather than reporting the first favorable numbers we found. Both rounds overturned or narrowed an initial claim: the "independent" LSTM evaluation was partly feature-family leakage (corrected macro F1 0.483, not 0.65, p=0.004), and the class-imbalance "fix" showed no significant evidence of improvement on retest (p=0.625), while the reported failure mode (focal loss) showed a strong but not conventionally significant signal (p=0.0625) that itself needs a larger seed count to confirm. We view this pattern of self-correction as a stronger contribution than either individual model result, and recommend it as a default practice for behavioral learning-analytics papers generally: report the entangled evaluation if that is what was run first, but do not present it as the headline number without a disjoint check; do not present a single-split comparison as a finding without a multi-seed significance test; and hold all p-values reported in a paper to the same significance threshold, including inconvenient ones. Future work should prioritize: (1) validating cluster-derived labels against a self-report instrument such as NASA-TLX (Hart & Staveland, 1988) on a smaller pilot sample, since clustering-plus-outcome-validation is still one step removed from ground truth, and would also directly address the construct-validity gap in Section 5.1; (2) a properly powered class-imbalance study (20+ seeds, per-comparison confidence intervals, and a wider set of techniques such as cost-sensitive sequence loss or SMOTE variants grounded in the broader imbalanced-learning literature (Chawla et al., 2002; He & Garcia, 2009) rather than jitter alone) before recommending any specific mitigation; (3) extending the same disjoint-evaluation and multi-seed rigor applied to Cognitive Load in Sections 4.5-4.6 to the Engagement target, which was flagged but not resolved in this paper (Section 5.1); (4) extending the LSTM to a true within-course time-varying prediction task using weekly-aggregated behavior rather than end-of-course labels, which would also weaken the residual label/input coupling noted in Section 4.5; and (5) testing whether a genuinely independent construct, such as a small pilot with real NASA-TLX self-report, correlates with either our entangled or strict-disjoint model output. This last direction connects naturally to adaptive intervention systems of the kind reviewed in Section 2.2, where a real-time load estimate could trigger micro-learning content restructuring, but only once the underlying detector's validity is established more rigorously than a single behavioral dataset permits, and only once the precision ceiling noted in Section 4.6 is meaningfully improved.

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